

Hypothesis Report Practice

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Abstract—This document provides responses to the short answer questions listed in the *Hypothesis Report Practice for CS-7641-O01*. It examines the paper “Tracking Everything Everywhere All at Once” by Wang et al., and its proposed algorithm, *OmniMotion*, a novel approach for estimating dense, long-range motion trajectories across entire video sequences. These responses address the authors’ backgrounds, the knowledge gap they filled, and the scope of ICCV within which their paper was recognized.

I. THE AUTHORS

As a start, who are the authors? How might they contribute to the larger field of paper’s topic and domain?

Seven authors, Qianqian Wang, Yen-Yu Chang, Ruojin Cai, Zhengqi Li, Bharath Hariharan, Aleksander Holynski, and Noah Snavely, contributed to “Track Everything Everywhere All at Once” [1]. Each author possesses an impressive background in computer science research, as a PhD student, university professor, or research scientist. Combined, they hold research experience with Cornell University, Google Research, or UC Berkeley, with plenty of additional projects and publications listed in their individual portfolios.

These seven authors are focused primarily on computer vision topics like 3D modeling and image analysis. For example, Noah Snavely, Ruojin Cai, Bharath Hariharan, and Qianqian Wang have together contributed to *Doppelgangers++*, a publication on their algorithm that differentiates visually similar but different 3D structures in separate images.

A brief review of any of their portfolios demonstrates his or her impressive ability in developing groundbreaking computer vision tools, systems, and algorithms. Aleksander Holynski aids in the development of one of the world’s most state-of-the-art image and video generation tools, Google Veo. Bharath Hariharan advises over twenty PhD students, where his former students have gone on to contribute their computer science knowledge at Fortune 500 companies.

Whether putting their effort into applied research, advising the next generation, or directly releasing new tools, each author has and continues to contribute to the computer vision domain.

II. ICCV

What is ICCV? What is the scope of the conference? How does this conference compare to the larger landscape of computer vision? Machine learning? You should provide context that gives broader understanding on how the paper might contribute to the field.

ICCV, or the International Conference on Computer Vision, is a global convention where attendees showcase papers, demonstrations, and discoveries in areas like image and video analysis, recognition, and generation. The conference presents the most cutting-edge work in computer vision to date using new or improved machine learning algorithms. “Track Everything Everywhere All at Once” by QianQian Wang et al. [1] was named the Best Student Paper, offering a profound impact on the field of image and video analysis.

While ICCV is arguably the most prestigious conference in computer vision, topics discussed typically make use of or develop new complex machine learning algorithms under the hood. Many accepted papers sometimes focus primarily on the underlying algorithms rather than higher-level use cases. For example, in 2015 the Marr Prize was awarded to “Deep Neural Decision Forests” by P. Kotschieder et al., which emphasized the use of decision trees for computer vision while introducing a novel classification approach combining them with deep neural networks – both classic machine learning models.

“Track Everything Everywhere All at Once” not only produced an impressively accurate method for tracking 3D movement in 2D clips, but also discussed its detailed use of RAFT, a neural network-based approach for predicting optical flow [1]. This demonstrates how, even though such research is renowned for its technological progression in computer vision, the focus often remains on advancing the machine learning techniques that are powering these steps.

III. THE GAP

Much like Rule 6 in Ten simple rules for structuring papers by Konrad Kording and Brett Mensh, what is the gap the authors provide?

The authors use the Introduction to familiarize the reader with current limitations of past tracking algorithms. While they provide brief context in the Abstract, it is explained further in the next section. The most common motion estimation approaches are contextualized, where the reader

is given a working knowledge before exposing their core constraint: producing dense and long-range pixel trajectories over extended periods of time [1]. Given this gap in the past algorithms, the authors break down their problem into 3 subcategories: tracking long sequences accurately, through occlusions, and coherently through space and time [1].

With an established issue, the new approach is introduced, checking off each subcategory with a brief description of its underlying mechanism. Finally, the section concludes with authors directly stating that their new approach solves the three established problems and substantially outperforms all other methods.

IV. HYPOTHESIS

What is the hypothesis of the article?

The authors begin their hypothesis in the Overview section, directly after the Introduction. It begins with their proposition of a new test-time optimization approach for estimating dense and long-range motion from a video sequence. They suggest that any pixel in any frame can be queried and tracked, resulting in a smooth and accurate travel path throughout the video.

V. ADDITIONAL RESOURCES

What specifics from the article do you personally need to look up for understanding the claims? How will this information provide context or focus? This can include background, notation, and citations.

I had to do some research on what *optical flow* was more specifically to better understand the article. Researching the term and learning how it is actually computed [3] gave me the best understanding of it as well as how it is computed. While it used plenty of context to help me infer what the term meant, there was no formal definition. Essentially, understanding the derived meaning of the term allowed me to better conceptualize the initial task of converting 2D images to 3D volumes and better rationalize what type of data the authors were working with.

I had a difficult time understanding the mathematical calculations, especially with an extensive number of variables and higher-level techniques. Reviewing the original article that covered these calculations [2] conceptualized them into more comprehensible ideas. It helped me understand how those functions represented dynamic scenes and how they were optimized with neural networks.

VI. LIMITATIONS

After reviewing and re-reviewing the article, what limitations do you discern? These can be simple or more complex observations.

A possible limitation I found would be identifying certain pixels upon reentry into the frame after an extended length of time. The authors cleverly used canonical volumes and bijective mappings to improve tracking accuracy and global consistency for pixels that remain visible in the video, but lose accuracy over time for pixels that completely escape the frame [1].

While their method performed well, reappearing points sometimes track slightly offset trajectories and slightly differing pixel regions. The method does not explicitly leverage appearance-based identity features, so large visual changes can cause ambiguity. Using an imaging recognition system to assign each pixel to a general pixel region may solve this problem, though large visual changes in pixel values may lead to further ambiguity.

VII. NEXT STEPS

What might be next steps to further this research? Think of a couple of different options. There are no correct answers here, however your answers must be grounded in reasoning.

Future steps considering this limitation may include implementing appearance-based re-identification to identify pixels that have escaped the frame, and therefore more accurately determine pixel trajectories.

When tracked points are occluded or out of frame, motion prediction becomes increasingly difficult over time [1]. Storing distinguishing visual features around the pixel would introduce a way to identify previous pixels when bijective mapping fails. These features could include sharp edges or high contrast between adjacent pixels, allowing the system to capture specific patterns and textures for re-identification when similar pixel regions appear.

Pixels within ambiguous regions (e.g. areas with uniform color) can rely on neighboring regions for motion inference, so even partial visibility could help reconstruct the region. Effectively, retaining non-spatial attributes allows for pixel identification regardless of its history or position within the frame.

Additionally, this method could integrate seamlessly into *OmniMotion* [1] as a refinement step. Instead of relying solely on regional recognition, it would fine-tune trajectories by focusing on a small search area around the predicted path. Rather than scanning the entire frame for matching regions, the algorithm would use the estimated trajectory as a reference point and limit the search to a defined radius. This would create a two-stage process that would first map trajectories onto a low-resolution volume, and then refine them to align with precise pixel patterns.

VIII. HYPOTHESIS

Please then provide a single hypothesis on what you'd like to investigate based on the knowledge gained in this exercise.

In addition to the *OmniMotion* system, incorporating appearance-based pixel re-identification by capturing and storing distinguishable pixel regions would improve estimated trajectories over extended periods of time. The current proposition made by the authors use coarse, low-resolution canonical volumes to track pixels, leading to drifting errors after long moments of occlusion. So incorporating region detection would refine a vague estimation to a far more precise pixel location.

REFERENCES

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